Power

power.kicad_sch

MCU

mcu.kicad_sch

Wireless

wireless.kicad_sch

IO

io.kicad_sch

RP2350 Wireless Dev Board — block overview

USB-C (5V) -> BQ24074 charger/power-path -> VSYS -> TLV75733P LDO -> 3V3
1-cell Li-Po on JST-PH (charged at 445 mA, VBAT/3 monitored on GPIO29)
RP2350A + 16 MB QSPI flash + 8 MB QSPI PSRAM (GPIO8 = XIP_CS1)
ESP32-C6-MINI-1 (Wi-Fi 6 / BLE 5) on UART1 with RTS/CTS flow control
microSD (SDIO 4-bit / SPI), Qwiic I2C0, WS2812B-2020 RGB LED, user LED

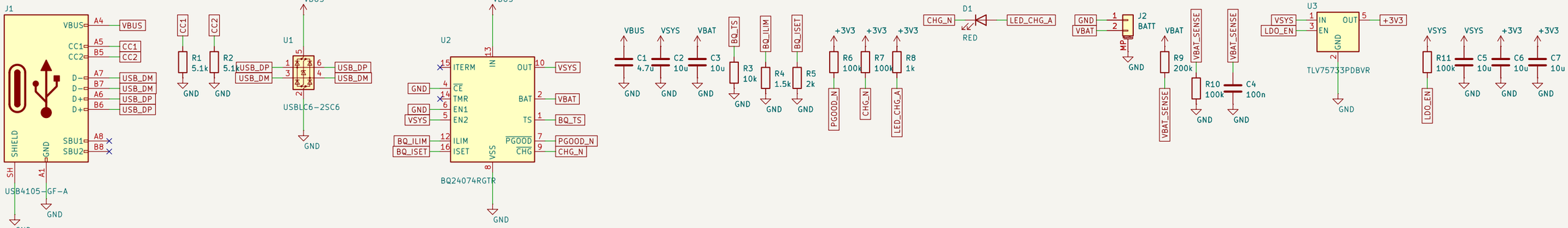
GPIO map
GP0/GP1  UART0 TX/RX (header)
GP2-GP5  header
GP6      ~PGOOD (USB power present, low)
GP7      ~CHG (charging, low)
GP8      PSRAM CS (XIP_CS1)
GP9      microSD card detect
GP10-15  microSD CLK, CMD, D0-D3
GP16/17  I2C0 SDA/SCL (Qwiic + header)
GP18     ESP32-C6 EN
GP19     ESP32-C6 IO9 (boot strap)
GP20-23  UART1 TX/RX/CTS/RTS to ESP32-C6
GP24     WS2812B data, GP25 user LED
GP26-28  ADC0-2 (header)
GP29     VBAT/3 (ADC3)

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**Title: RP2350 Wireless Dev Board — Top level**

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USB−C sink: 5.1k Rd on CC1/CC2 (default USB power).
BQ24074 power−path: OUT (VSYS) is fed from VBUS when present,
battery supplements via DPPM. EN2=1/EN1=0 −> ILIM resistor mode,
R4 1.5k −> 1.03 A input limit. R5 2k −> 445 mA charge current.
TMR open = default safety timers, ITERM open = 10% termination.
TS: 10k fixed resistor (packs without NTC).

VBAT sense: VBAT/3 on GPIO29 (ADC3), same scaling as Pico 2.
Divider draws ~14 uA from the cell.
3V3: TLV75733P 1 A LDO from VSYS; EN on header (pull low = off).
Battery: JST−PH 2 pin, pin 1 = GND, pin 2 = +VBAT (Adafruit/SparkFun
polarity). CHECK YOUR CELL'S POLARITY BEFORE CONNECTING.

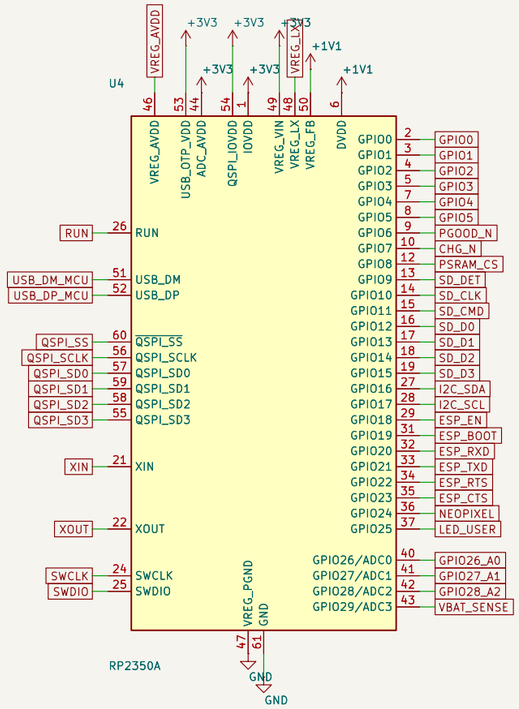


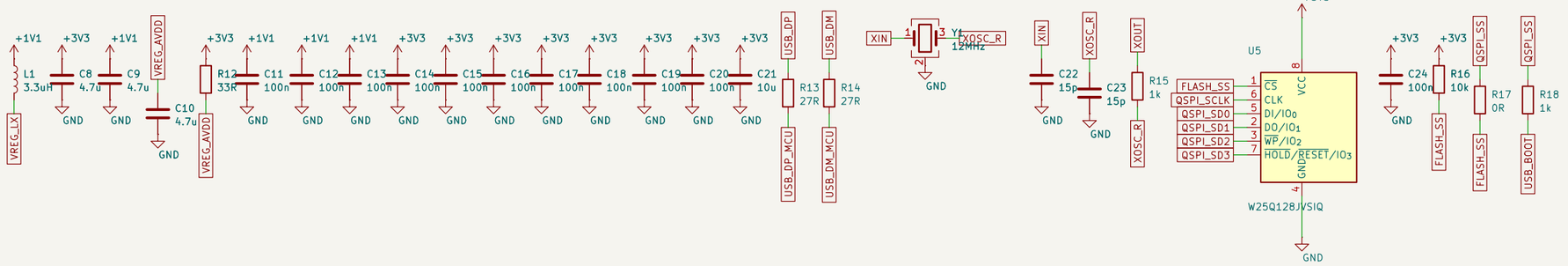

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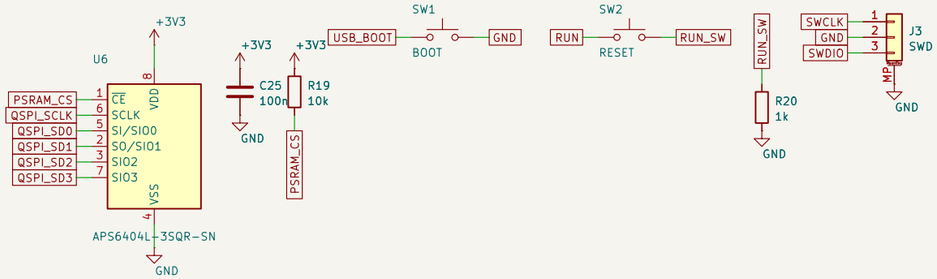
Sheet: /Power/
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**Title: RP2350 Wireless Dev Board − Power: USB−C, Li−Po charger / power−**

| Size: A3 | Date: 2026−11−09 | **Rev: A** |
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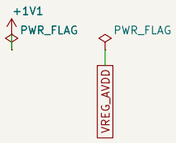



RP2350A core copied from Raspberry Pi 'RP2350A Minimal' R4 reference (MIT licence): regulator (L1 polarity-marked AOTA-B201610, C8/C9/C10, R12 33R), crystal (ABM8-272-T3, 15p, 1k), decoupling, flash, BOOTSEL. Schematic refs differ; PCB placement/routing of this block is the RPi layout rotated 90 deg.

PSRAM (APS6404L, 8 MB) on XIP_CS1 = GPIO8 (same as Adafruit Feather RP2350 so existing board configs work). Flash 16 MB on QSPI_SS.
BOOT button doubles as a user button (read via QSPI_SS).
SWD on JST-SH 3 pin (Raspberry Pi Debug Probe pinout).

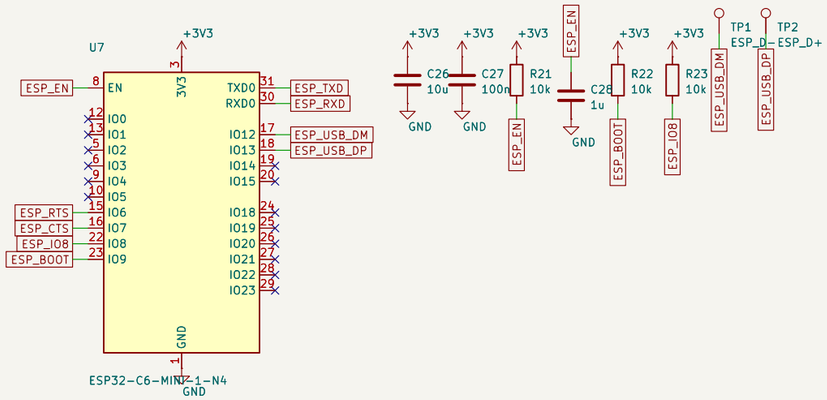



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Sheet: /MCU/
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**Title: RP2350 Wireless Dev Board — RP2350A core, QSPI flash, PSRAM**

| Size: A3 | Date: 2026-11-09 | Rev: A |
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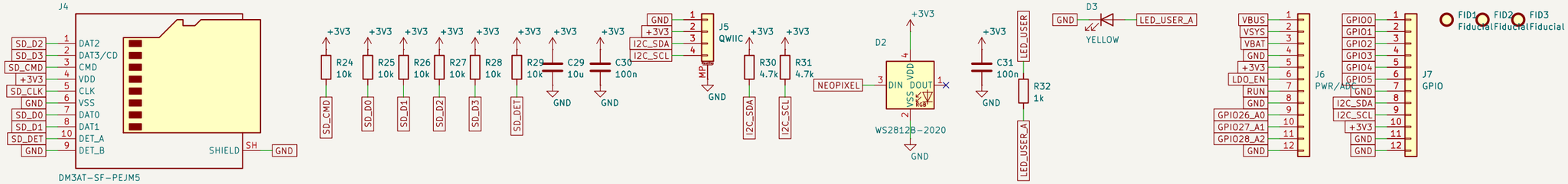
ESP32-C6-MINI-1 runs ESP-AT or ESP-Hosted firmware as a Wi-Fi 6 / BLE 5 co-processor. Link: RP2350 UART1 (GPIO20 TX, 21 RX, 22 CTS, 23 RTS) <-> ESP UART0 (RXD0/TXD0) + IO6 (RTS) / IO7 (CTS). GPIO18 = ESP_EN (drive low to reset), GPIO19 = ESP IO9 strap (low at reset = ROM download mode) -> RP2350 can re-flash the ESP over the same UART (esp-serial-flasher). IO8 pulled up (required for download mode). ESP native USB on test pads TP1/TP2.

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***Title: RP2350 Wireless Dev Board — ESP32-C6 Wi-Fi 6 / BLE 5 co-process***

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microSD wired for 4-bit SDIO via PIO (CLK GPIO10, CMD GPIO11,
D0-D3 GPIO12-15 consecutive) and also usable in SPI mode
(SPI1: SCK=10, MOSI=11, MISO=12, CS=15). Card detect GPIO9 (low = card).
Qwiic/STEMMA QT on I2C0 GPIO16/17 with 4.7k pull-ups.
WS2812B-2020 on GPIO24 (powered from 3V3). User LED GPIO25.

Headers are 2.54 mm, fitted on the bottom side (breadboard use).
J6: VBUS VSYS VBAT GND 3V3 EN RUN GND A0 A1 A2 GND
J7: GP0(TX) GP1(RX) GP2 GP3 GP4 GP5 GND SDA SCL 3V3 GND GND

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Sheet: /IO/
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***Title: RP2350 Wireless Dev Board — microSD, Qwiic, RGB LED, headers***

| Size: A3 | Date: 2026-11-09 | **Rev: A** |
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| KiCad E.D.A. 10.99.0-unknown | | Id: 5/5 |